A 5V power bank is used to power the Raspberry Pi.

Navigation. The navigation circuit controls movement of the robot. It consists of Arduino Mega 2560 board (Board2), HC-05 Bluetooth module, L293D motor driver (IC1), and two DC geared motors M3 and M4.

Directional data or command from the smartphone is sent to the navigation circuit through HC-05 Bluetooth module. It is processed by the Arduino and then fed to the motor driver IC, which drives the geared motor in the required direction. Commands are given to the Bluetooth Electronics app installed in the smartphone.

First, you need to open the Bluetooth Electronics app and pair with the HC-05 module as shown in Fig. 3. Once the two devices are paired, the buttons are edited in the app and configured with English alphabet characters. Each direction (forward, backward, right and left) is assigned a character. Four buttons are used as shown in Fig. 4. When these buttons are pressed, the corresponding characters get transmitted.

HC-05 Bluetooth module receives the characters and sends these to the Arduino pins (transmitter and receiver). The Arduino processes this data and compares with the Arduino program. The corresponding digital values are sent to input pins of L293D IC. The driver IC provides more current in order to make the motor move in the required direction. Depending on the data received from the Bluetooth module, the motor moves in forward or backward direction.

There are four wires connected to the geared motors from Board2 having ATmega2560 microcontroller (MCU).

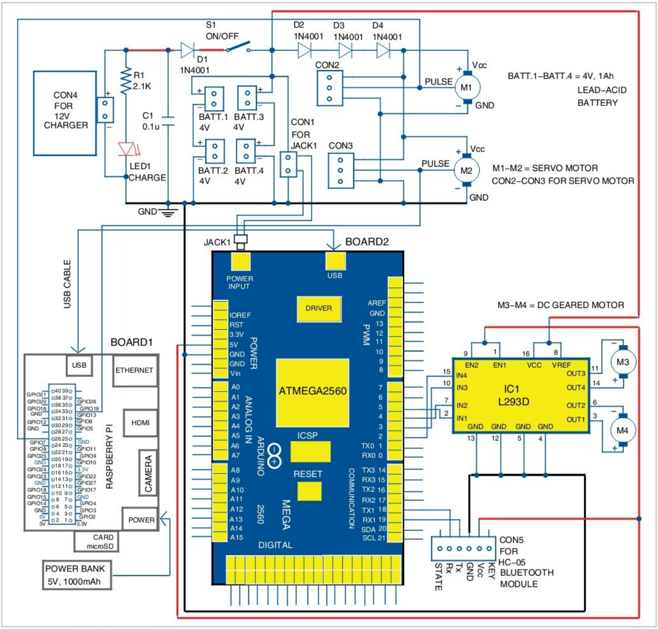

Fig. 2: Circuit diagram for virtual telepresence robot